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MedGraph: Maintaining Patient State with Graph Memory for Long-Term Medical Dialogue

Three-Layer Hierarchical Graph Memory Eliminates Outdated State, Cross-Patient Contamination, and Missing Safety Facts, Gaining 10.5 Points on MedMemoryBench

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Medical dialogue assistants fail in longitudinal settings because flat memory systems accumulate outdated states, mix facts across patients, and silently drop safety information. MedGraph introduces a three-layer hierarchical graph memory that makes clinical state transitions explicit and unconditionally prepends safety facts to every query. On MedMemoryBench it exceeds the strongest of five baselines by 10.4–10.5 percentage points with both GPT and Qwen backbones.

AT A GLANCE

Problem: Flat memory systems for clinical dialogue produce outdated state, cross-patient contamination, and silently missed safety facts across multi-visit conversations.

Why it matters: LLMs used as clinical assistants in longitudinal care must maintain accurate, up-to-date patient state; errors can cause harmful medication or diagnosis decisions.

Method: Three-layer memory: raw transcripts (L1), a typed clinical fact graph with temporal validity tags (L2), and a patient profile unconditionally attached to every query (L3).

Result: 10.4–10.5 pp above the strongest of five baselines on MedMemoryBench with GPT and Qwen backbones.

Evidence: MedMemoryBench (multi-visit clinical dialogues); baselines: flat vector, sliding window, RAG, Zep, MemoryOS.

The Big Picture

Large language models deployed as clinical dialogue assistants need to maintain accurate knowledge of a patient's medical state across many visits over weeks or months. Existing memory approaches — flat vector stores, sliding-window context, retrieval-augmented generation — treat memory as a bag of text fragments. This fails in three recurring ways identified by the authors through systematic error analysis of five baseline systems on longitudinal medical dialogues:

1. **Outdated state:** A resolved symptom is reported as currently active because an older transcript fragment scores higher in retrieval than the later discharge note.
2. **Cross-patient contamination:** Facts from a different patient's earlier session pollute the current patient's context when embedding similarity overrides patient identity.
3. **Missing safety facts:** A known allergy or contraindication is not retrieved because it was not in the top-k most similar fragments for the current query.

MedGraph introduces structure that makes all three failures impossible by design: typed temporal graph updates prevent outdated state; strict per-patient graph isolation prevents contamination; and mandatory safety-block attachment prevents silent omission.

Why Existing Approaches Fall Short

Flat vector memory embeds all clinical utterances and retrieves the top-k most similar. It does not distinguish whether a fact is currently active or resolved, and similarity can produce facts from other patients if patients share medical vocabulary.

Sliding-window context keeps the most recent n tokens. Long-term facts (a years-old allergy, a chronic condition diagnosed at first visit) fall out of the window and are lost.

Retrieval-augmented generation (RAG) retrieves relevant documents at query time. It improves recall of historical facts but cannot distinguish the current clinical state from historical states when multiple conflicting observations exist in the corpus.

Zep (entity memory) maintains entity states but uses flat entity records without typed temporal transitions; it conflates active and resolved states.

MemoryOS provides structured memory with hierarchical layers but does not specialize for clinical entity types or enforce mandatory safety-fact prepending.

Core Mechanism: Three-Layer Memory

Layer 1 — Raw dialogues. Full session transcripts are stored verbatim, enabling exact retrieval of clinical reasoning chains and direct-quotation evidence when needed. This layer is searched by traditional BM25 retrieval for specific historical passages.

Layer 2 — Clinical fact graph. A typed knowledge graph over entities (symptoms, diagnoses, medications, lab results, procedures) with typed relations (has-symptom, prescribed, diagnosed-with, resolved, contraindicated-with). Each entity node carries a temporal validity tag: *active*, *resolved*, or *historical*. Graph construction and update are performed by a clinical NLP extraction pipeline applied to each new session.

Layer 3 — Patient profile (safety block). A compact structured record per patient containing: recorded allergies, contraindications, chronic conditions, emergency contacts, and medication alerts. This block is *unconditionally prepended* to every query context, bypassing the retrieval mechanism entirely. No fact in this block can be missed because retrieval does not decide whether it is included.

Graph isolation. Each patient has a fully separate graph instance and a separate patient profile. Cross-patient contamination is architecturally impossible.

Technical Formulation

Let $\mathcal{G}_p = (V_p, E_p)$ be the clinical fact graph for patient p, where each node $v \in V_p$ is a clinical entity with attributes $(\text{type}, \text{value}, \text{validity})$ and each edge $e = (v_i, v_j, r)$ is a typed relation.

When a new session transcript D_t arrives for patient p, the extraction pipeline produces a set of entity mentions $\hat{V}_t$ and relation mentions $\hat{E}_t$. The update procedure applies type-specific rules:

$$\text{if } v \in \hat{V}_t \text{ and } v \notin V_p: \quad \text{add } v \text{ with validity} = \textit{active},$$

$$\text{if } v \in \hat{V}_t \text{ and } v \in V_p \text{ and ``resolved'' signal:} \quad \text{set validity} \rightarrow \textit{resolved},$$

$$\text{if } v \in \hat{V}_t \text{ and } v \in V_p \text{ and no state change:} \quad \text{update timestamp only}.$$

The "resolved" signal is detected by a small classifier trained on clinical resolution phrases ("symptom resolved", "discontinued", "normal finding", etc.).

At query time, the context assembled for the LLM backbone is:

$$\mathcal{C} = \underbrace{\text{Profile}_p}_{\text{L3, mandatory}} \oplus \underbrace{\text{Graph}_p^{\text{active}}}_{\text{L2, active facts}} \oplus \underbrace{\text{RAG}(\mathcal{D}, q)}_{\text{L1, retrieved}},$$

where $\text{Graph}_p^{\text{active}}$ is the subgraph of currently active entities serialized as structured text, and $\text{RAG}(\mathcal{D}, q)$ is BM25-retrieved transcript fragments relevant to query q.

Learning or Inference Procedure

MedGraph uses no end-to-end learning. Its components are:

- A clinical NER/RE pipeline (pre-trained, fine-tuned on MIMIC-IV notes) for entity and relation extraction.
- A resolution-signal classifier (a small BERT-based model fine-tuned on clinical resolution annotation).
- An arbitrary LLM backbone (tested with GPT-4o-mini and Qwen2.5-72B) for the final answer generation.

The NLP pipeline and classifier are frozen at inference time.

What the Guarantee Says

No formal guarantee is claimed. The empirical claim is superior clinical accuracy across all five baselines on MedMemoryBench with both LLM backbones. The error analysis identifies three failure modes and shows that MedGraph's design eliminates each by construction; the reduction in those specific error types is verified per-category.

Experimental Findings

- **Overall accuracy (GPT-4o-mini):** MedGraph +10.5 pp above best baseline (Zep).
- **Overall accuracy (Qwen2.5-72B):** MedGraph +10.4 pp above best baseline.
- **Outdated-state errors:** Reduced by 68% relative to Zep, confirming typed temporal update effectiveness.
- **Safety-fact recall:** 100% safety-fact recall (by mandatory prepending), vs. 73–82% for retrieval-based baselines.
- **Cross-patient contamination rate:** 0% (by graph isolation), vs. 3–7% for vector-memory baselines.

Ablations and Interpretation

Without mandatory safety block: Dropping the unconditional prepend and relying on retrieval for safety facts reduces safety-fact recall from 100% to 79%, confirming that retrieval alone is insufficient.

Without typed temporal updates: Using simple entity overwrite (new mention always replaces old) increases outdated-state errors by 3.2×, as resolved symptoms return when re-mentioned in a summary context.

Without graph isolation: Merging all patients into a single graph increases cross-patient contamination to 4.1%, confirming that per-patient isolation is necessary.

L1 retrieval necessity: Removing L1 (raw transcript retrieval) reduces overall accuracy by 4.3 pp, showing that verbatim historical passages provide useful evidence beyond what the structured graph captures.

Reference

Donghua Zhang, Zizheng Wang, Ze Dong, Yiding Feng, Siyu Qian. **MedGraph: Maintaining Patient State**

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